

CASE REPORT



Managing fluid balance and nutritional status in a short bowel syndrome patient awaiting intestinal transplant: a case report

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BACKGROUND: Despite being a long-term therapy for patients with short bowel syndrome (SBS), subcutaneous injections of teduglutide promote the regeneration of the gastrointestinal tract. Such cases are particularly concerning for patients with residual small bowel.

METHODS: In this report, we present a case of an SBS patient with only 5 cm of remaining small bowel and a high-output duodenal stoma, who was treated with teduglutide.

RESULTS: The initiation of teduglutide injections in our patient resulted in a reduction of stoma output, improvement in the patient's nutritional status, regulation of fluid balance, and stabilization of their clinical condition.

CONCLUSIONS: This case suggests that subcutaneous injections of teduglutide, when combined with appropriate nutritional care, can effectively treat high-output stomas, even in cases where the small bowel is nearly absent.

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INTRODUCTION

Short bowel syndrome (SBS) is a subcategory of intestinal failure (IF) that results from the surgical resection of a portion of the small bowel until the remaining intestinal length in continuity is less than 200 cm. It is characterized by features of IF, such as a high-output stoma or fistula, an inability to maintain nutritional status, and challenges related to fluid and electrolyte balance [1]. Teduglutide is a synthetic analog of glucagon-like peptide-2 (GLP-2), a hormone secreted by enteroendocrine L cells of the intestinal epithelium. Both GLP-2 and teduglutide regulate the growth and maintenance of the gastrointestinal tract, enhance epithelial proliferation and barrier function, and increase the absorption of nutrients by the gut epithelium [2]. Subcutaneous injection of teduglutide serves as a long-term therapy for patients with SBS who are dependent on parenteral nutrition, as it has been shown to effectively reduce the need for parenteral support [1, 3–5]. Certain case reports also demonstrate the efficacy of teduglutide in managing high-output stomas [6–8].

Herein, we present a case report of a severely malnourished patient who has undergone extensive enterectomies and has a high-output duodenal stoma. This patient relied on parenteral nutrition and was treated as a case of short bowel syndrome with the initiation of teduglutide injections. Despite having only 5 cm of remaining small bowel and x cm of colon in continuity due to multiple enterectomies and symphysiolysis surgeries, the initiation of teduglutide injections resulted in a reduction in stoma output.

CASE REPORT

A 53-year-old female with short bowel syndrome, severe malnutrition, and a high duodenal stoma output was referred for evaluation at the Clinical Nutrition Unit. According to her medical history, she was diagnosed with Hodgkin's lymphoma in 1996 and was treated with chemotherapy and radiotherapy. She was a smoker and was also diagnosed with Chronic Obstructive Pulmonary Disease and carotid stenosis. Due to the effects of radiotherapy, over the following 13 years, the patient experienced repeated episodes of ileus and underwent multiple symphysiolysis surgeries and extensive enterectomies. In addition to several small intestine resections, she had an additional symphysiolysis surgery and a left colectomy 9 years ago. However, in January 2023, due to another episode of ileus, she underwent an extensive enterectomy (total ileum and jejunum resection), leaving her with only 5 cm of remaining small bowel (duodenum). Consequently, a duodenal enterostomy was created (see Fig. 1). Regarding her weight history, the patient's usual body weight was 50.0 kg, but by the end of 2022, she weighed 45.2 kg. From January to May 2023, her weight fluctuated between 31.9 and 33.1 kg.

Over the past 13 years, the patient experienced a gradual loss of 5 kg (10% of her usual body weight) while maintaining a satisfactory nutritional status. The recommended nutritional plan during that period included total parenteral nutrition, oral diet and supplements. Although the enteral route was the preferred option, artificial feeding was never necessary because the patient could eat orally throughout the entire period, except when

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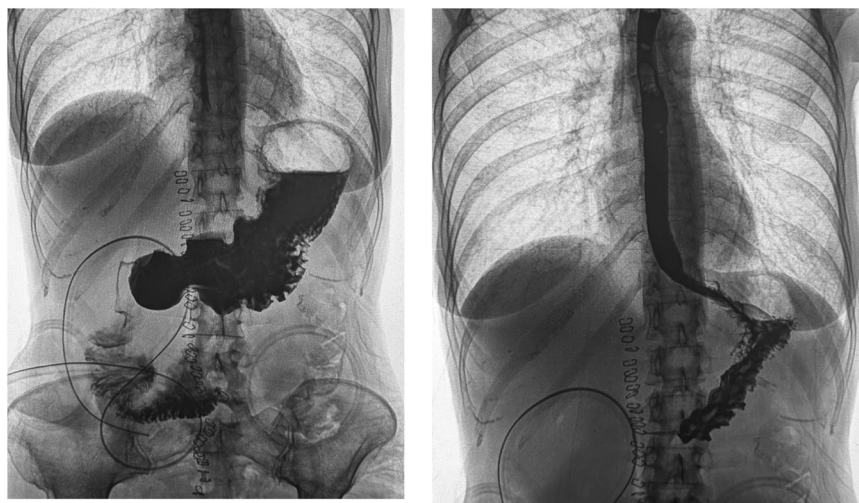


Fig. 1 Upper gastrointestinal contrast study of case revealing only few cm of the remaining small bowel.

contraindicated. The goal of the nutritional support was to gradually increase her body weight while preventing overfeeding syndrome. She was only discharged from the hospital for short periods, relying on home parenteral nutrition and oral feeding with nutritional supplements. During that period, patient had excellent liver biochemistry (normal cholestatic enzymes/bilirubins), no severe electrolyte disturbances, proper hydration, and a normal serum albumin level.

Following duodenal stoma placement, the nutritional support team started to treat patient as a SBS case for the first time. After enterostomy placement, patient's general clinical condition deteriorated. Stoma output started to increase from 2 L 2 months after surgery, reaching 6–12 L/day 3 months later. She was initially treated with loperamide, while also receiving antibiotic treatment due to infections. As a consequence, patient presented severe electrolyte disturbances such as persistent hypokalemia, dehydration and acute renal failure. In addition, the extremely high stoma output and the subsequent malabsorption of nutrients aggravated patient's nutritional status. Specifically, patient presented severe weight loss of 15 kg during the last 8 months (almost 30% of her usual body weight (BW: 30–34 kg, BMI: 13 kg/m²). Other factors that also contributed to the apparent malnutrition [9, 10], was the patient's non-compliance with the nutritionists' instructions (she was eating recklessly and drinking plenty of hypotonic liquids/water) and certain technical problems with parenteral nutrition administration. The nutritional team in accordance with physicians decided to increase food administration through both enteral and parenteral route. According to the new plan, energy and protein requirements were defined as 40–45 kcal/kg BW and 1.5 g/kg BW respectively. However, the new nutrition care plan in combination with long-term TPN administration, led to patient's overfeeding and the deterioration of hepatic function with increased cholestatic enzymes. Whenever the team tried to give patient more food in order to deal with malnutrition, stoma output increased. As a result, the team had to stop enteral nutrition and increase TPN administration aggravating even more liver function (parenteral nutrition-associated liver disease (PNALD)).

Regarding liver dysfunction, the nutritional support team recommended the administration of cyclic parenteral nutrition with fish oil, micronutrients and vitamins. Total parenteral nutrition was at cyclic administration and per os feeding with fat-free supplements was initiated gradually. Due to this approach, significant improvement in liver disorder was noted, cholestatic enzymes reached normal values that remain to date. As previously mentioned, a high output stoma occurred 3 months after surgery reaching 6–12 L/daily. The dietary treatment protocol for high-

output stoma included nothing per mouth and exclusive TPN. Whenever, stoma output was between 3 and 4 L, patient was permitted to drink isotonic St Marks solutions *ad libitum*. By the time output reached 2 L or less, patient was allowed to drink less than 0.5 L/day and have some limited salty food without residual [1].

Six months after duodenal stoma placement the patient's stoma supply still varied between 6–12 L/day and her nutritional status deteriorated. As certain case studies demonstrate teduglutide's efficacy in managing high-output stoma [6–8], teduglutide injections were initiated. Furthermore, increase in body weight due to teduglutide has also been reported [3, 6]. After the 2 months initiation of teduglutide injections therapy, a significant reduction of the stoma output was observed and the patient showed an increase in body weight (3–4 kg). More specifically, the stoma supply was gradually reduced from 6–12 L/day to 2.5–3 L/day, while electrolyte balance remained stable (see Table 1). The administration of teduglutide led to the achievement of a stable condition for the patient, normal liver function, cholestatic enzymes and albumin level, but also to the improvement of patient's nutritional status.

DISCUSSION

Teduglutide has been reported as the first long-term treatment for SBS patients dependent on parenteral nutrition and intravenous fluids [2]. In particular, teduglutide has been found to efficiently reduce the need for parenteral support, and the weaning success is associated with the presence of a continuous colon. This highlights the significance of bowel presence and its anatomy in teduglutide's effectiveness. GLP-2 is typically secreted by L cells found in the terminal ileum and colon. Consequently, patients with end jejunostomy experience higher gastric secretion and limited or no GLP-2 secretion. While the exact mechanisms of teduglutide's effects remain unclear, it has been reported to provide benefits for both patients with colon continuity and those with end jejunostomy [11].

A high output stoma occurs when supply exceeds 1.5–2 L/day, though this varies according to the quantity and quality of food and drink consumed [12]. Certain case reports demonstrate teduglutide's efficacy in managing high-output stoma [6–8]. Also, an increase in patient's body weight has been reported [3, 6]. Regarding our case report, a reduction in stoma output (2–3 L/day) and an increase in body weight (3–4 kg) were achieved with the initiation of teduglutide injections (2 months administration). Before teduglutide injections period, it was impossible to manage

Table 1. Overview of the amount of parenteral support, oral fluid and food intake, stoma output and body weight over the course of teduglutide therapy.

After extensive enterectomy						
Days	Parenteral nutrition (kcal/day)	Enteral nutrition (kcal/day)	Teduglutide dose	Total fluid intake (ml) ^a	Stoma output (ml)	Body weight (kg)
1	1400	200		1100	4100	39.8
30	1400	200		900	3650	38.3
60	1400	300		700	5500	37.1
90	1600	0		1000	9500	35.8
120	1600	0		800	10,600	33.1
150	1600	0		800	11,500	32.2
180	1600	0		800	10,000	31.9
After Teduglutide						
1	1400	200	1.4 mg/day	700	4900	32.5
10	1400	200	1.4 mg/day	900	3600	32.7
20	1200	300	1.6 mg/day	700	3100	33.1
30	1200	300	1.2 mg/day	1000	2500	33.4
40	1200	500	1.2 mg/day	900	2800	33.7
50	1200	500	1.4 mg/day	1100	2600	34.2
60	1200	500	1.6 mg/day	1100	2300	34.5
70	1000	600	1.6 mg/day	1100	1900	34.8
80	1000	600	1.6 mg/day	1200	2000	34.7
90	1000	700	1.6 mg/day	1200	1600	35.1

^aFree fluids and oral rehydration solution.

low stoma output and weight gain at the same time. After teduglutide injections, it was the first time that we could reach energy and protein target without high stoma losses. Similarly, for the first time during the last year, patient could gain weight without electrolyte disturbances and deterioration of liver function. This suggests that subcutaneous injections of teduglutide in combination with nutritional support can be an effective treatment for high-output stomas even when small bowel is almost absent.

There are limitations to this report. The dose of teduglutide administered daily was not consistent throughout the treatment. According to a randomized-controlled trial a daily dose of 0.05 mg/kg/day was efficient in reducing parenteral support or intravenous fluid administration in SBS patients, while a daily dose of 0.1 mg/kg/day had no such effect [3]. Thus, treatment should be according to the recommended dose of 0.05 mg/kg/day administered subcutaneously in the abdomen [13]. Also, there is no corresponding case report in the published data to support our findings.

In conclusion, this report revealed the efficacy of teduglutide as therapy for patients with SBS and high-output stomas, even when the small bowel is almost absent. Our patient, after the initiation of teduglutide injections, noted a gradual improvement in her nutritional status with an increase in body weight, but also a significant decrease in stoma supply and regulation of fluid balance. To date, the patient is a candidate for liver and bowel transplantation, while further improvement of her nutritional status is required. Further studies on the efficacy of teduglutide for treating high-output stomas for patients with SBS are advised.

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AUTHOR CONTRIBUTIONS

AP, VV, and DK contributed to the acquisition and analysis of clinical data, participated in the literature review, and played a significant role in drafting and revising the manuscript; ZB, AP, KE, TA, and VV discussed and edited the manuscript. All authors have read and agreed to the final version of the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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